EXPERIMENT - 23

23. Implement the Top hat technique as a Morphological operation to dilate the foreground regions based on Open CV.

Aim

To implement the Top Hat technique as a morphological operation using OpenCV in Python to highlight small bright regions in an image.

Procedure

1. Import the OpenCV and NumPy libraries.
2. Read the input image using cv2.imread().
3. Resize the image to fit the screen.
4. Convert the image into grayscale.
5. Create a kernel using np.ones().
6. Apply the Top Hat operation using cv2.morphologyEx() with cv2.MORPH_TOPHAT.
7. Display the original image and the Top Hat output image.

Program

```
import cv2
import numpy as np

img = cv2.imread("image.jpg")

if img is None:
   print("Image not found")
else:
   img = cv2.resize(img, (500, 350))
   gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

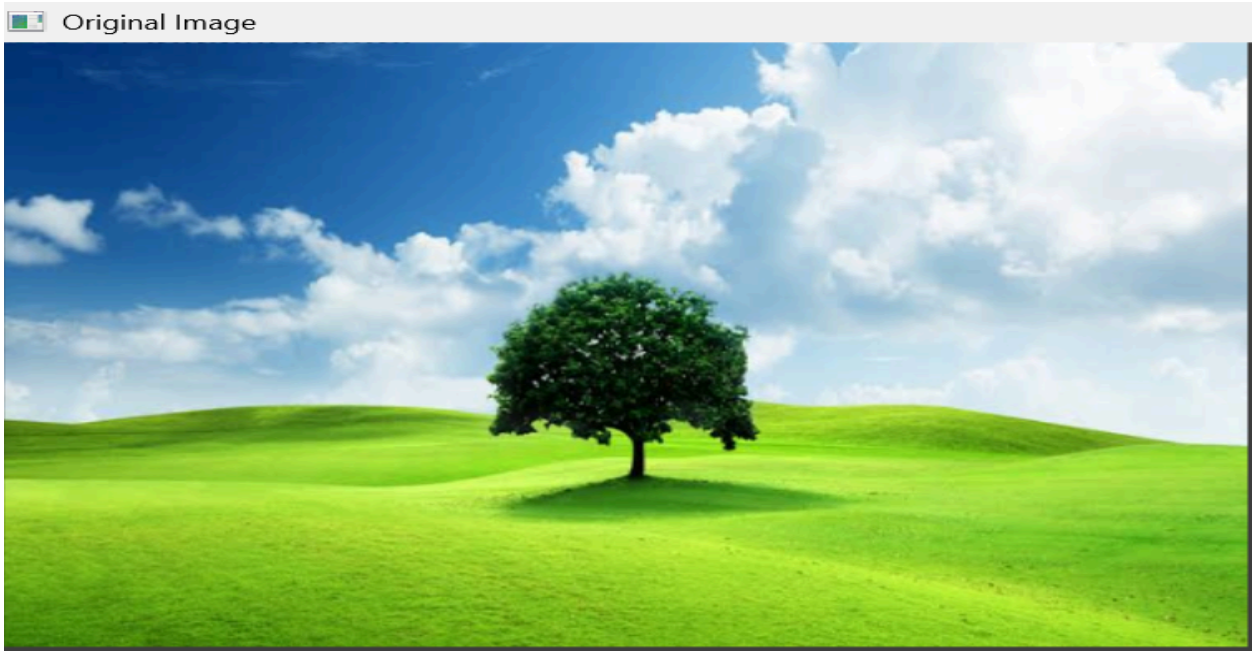
   kernel = np.ones((5, 5), np.uint8)

   tophat = cv2.morphologyEx(gray, cv2.MORPH_TOPHAT, kernel)

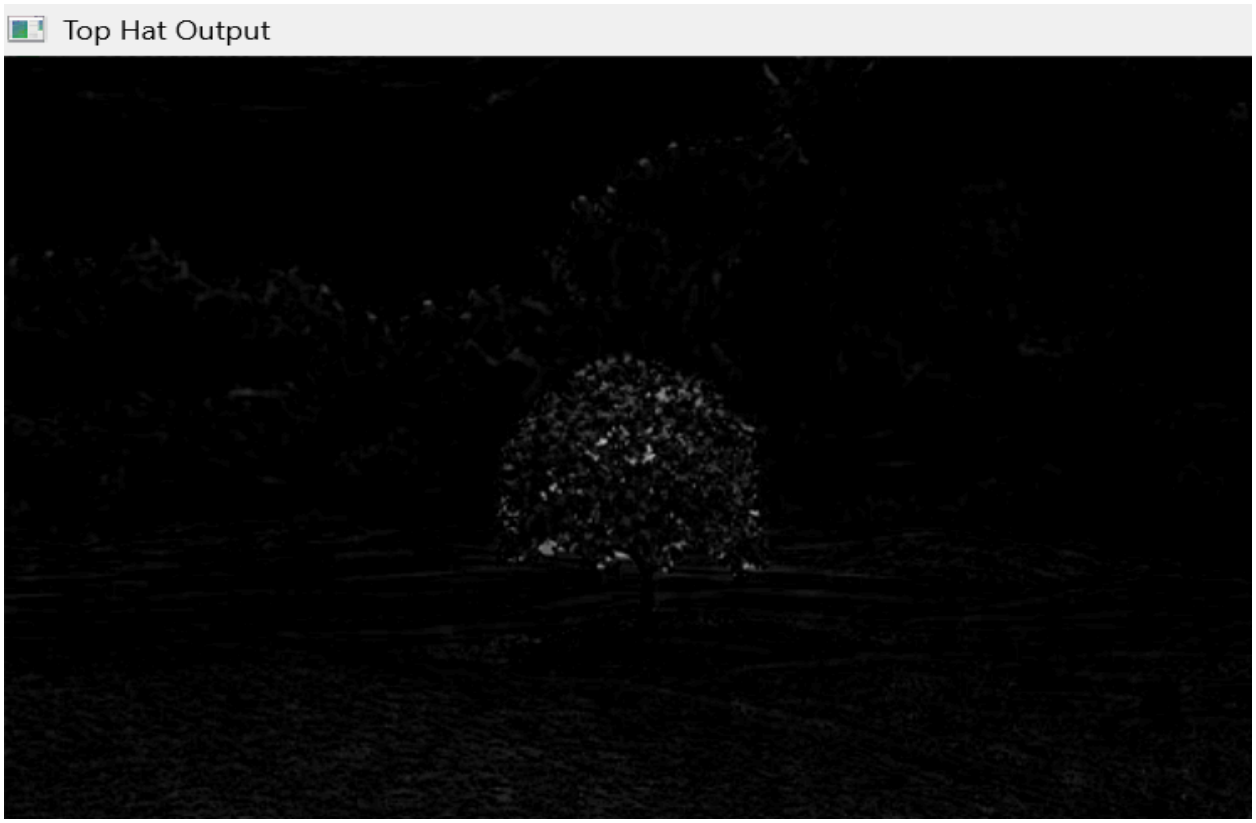
   cv2.imshow("Original Image", img)
   cv2.imshow("Top Hat Output", tophat)

   cv2.waitKey(0)
   cv2.destroyAllWindows()
```

Input Image



Output Image



Result

The Top Hat Morphological Operation was successfully implemented using OpenCV in Python.